Kaustubh Gupta

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Research Interests

- **Areas:** AI safety and alignment, AI welfare, interpretability, LLM evaluation (multilingual, agentic)
- **Methods:** empirical ML, mechanistic interpretability and steering, behavioural evaluation, LLM-judge calibration, preference elicitation, statistical evaluation

Education

Indian Institute of Technology Hyderabad, India, B.Tech in AI and Engineering Science — 2025
GPA: 9.34/10.0; Class Rank: 1/34

- **Relevant coursework:** Probability, Convex Optimization, Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Speech Systems, AI for Humanity

Experience and Fellowships

Oracle, Data Scientist 1 (mentor: **Karthik Bangalore Mani**) — Bangalore, IN
July 2025 – present

- Ship LLM query understanding for FAW Copilot: entity classification, column reranking, RAG.
- Improve Copilot quality via prompt tuning, JSON response contracts, backend-log analysis.

CultureSEEDS, Research Collaborator — Remote
Oct 2026 – present

- Contribute to literature review for a Google DeepMind-led survey on multilingual AI evaluation.

Macquarie University, Research Collaborator — Remote
Sept 2026 – present

- Co-develop multilingual LLM refusal evaluation with **Prof. Usman Naseem** and **Utsav Maskey**.
- Built 15-language, 8-task refusal dataset (211k prompts, on Hugging Face); piloting 4 models.
- Replicated English task-over-refusal benchmark (Llama-3.1-8B, n=270) with a local gpt-oss:20b judge.

Sentient Futures, Research Fellow — Remote
Aug 2026 – present

- Lead persona-vector steering workstream (mentors: **John Lund**, AE Studio, CaML).
- Built Qwen3-8B steering and fine-tuning-erosion pipeline: layer sweep, dynamic steering schedule, judge validation against human labels; main run in progress.

Better Futures Fellowship, Research Fellow — Remote
Aug 2026 – Oct 2026

- Tested engagement prompts for AI takeover of user decisions (code): 5 open-weight models, 8 scenarios, 720 trajectories, 2 LLM judges.
- Engagement prompts raised takeover 23 to 25 pp (95% CI [11, 39]) vs neutral; autonomy prompts cut it 32 to 33 pp.

Oracle, Data Analytics Intern — Bangalore, IN
May 2024 – July 2024

- Built LLM synthetic-data pipeline for FAW (K-Means, farthest-point sampling).

Honors and Awards

- Department highest CGPA, B.Tech in AI and Engineering Science, IIT Hyderabad (2025).
- Top 100, Meta × PyTorch × Hugging Face OpenEnv Hackathon (31,000+ teams), Apr 2026.

Publications

Interpreting and Steering State-Space Models via Activation Subspace Bottlenecks — ICML 2026

- Vamshi Sunku Mohan, **Kaustubh Gupta**, Aneesha Das, Chandan Singh.
- Steering activation subspace bottlenecks improves 7 Mamba-family SSMs on 6 benchmarks (+8.27% on average).
- Role: ran perturbation and steering experiments on delta neurons; contributed to experiments and paper writing.

Stream-TTS: A Low-Latency Text-to-Speech using Kolmogorov-Arnold Networks for Streaming Speech Applications — IEEE ICASSP 2025

- Giridhar Pamisetty, Riya Ann Easow, **Kaustubh Gupta**, Prof. K. Sri Rama Murty.
- Multilingual streaming TTS with Kolmogorov-Arnold Networks: 0.0795 RTF, 4.09 MOS.
- Role: designed and trained the base model architecture (EE6307 Speech Systems course project).

Fairness-Aware Submodular Optimization: A Survey — Unpublished

- **Kaustubh Gupta**, supervised by Prof. Aravind Reddy.
- Survey of fairness-aware submodular maximization: organized formulations by constraint type (cardinality, matroid, knapsack), separated discrete from continuous (multilinear-extension) algorithms, and compared their fairness–utility tradeoffs.
- Role: did the literature search, formulation comparison and writing end-to-end (EE 5328 Introduction to Submodular Functions, Jul–Nov 2024).

Attentive Multimodal Learning for Suicide Risk Assessment Using Acoustic and Prosodic Cues — Under review

- Giridhar Pamisetty, **Kaustubh Gupta**, Prof. K. Sri Rama Murty.
- Role: built the initial acoustic-embedding and prosodic-cue pipeline.

Projects

PrefKit: Multi-Method Preference Elicitation Toolkit — Apart Digital Minds

- Built 4-method elicitation toolkit with Cross-Method Score; ran Qwen3 1.7B–14B on 24 outcomes.

Noticing the Move, Missing the Player: LLM Agents Detect Behavioral Shifts but Not Peer Swaps — Apart AI Collusion

- Tested peer-swap detection in 3 multi-agent settings; behavioural-probe AUC 0.915–0.984.

Detection Is Not Containment: Timely Monitoring of AI Agent Failures — Apart Research

- Co-built incident claim-extraction pipeline; detection helps only if a last-safe stop point remains.

Optical Character Recognition for Ancient Sanskrit Manuscripts — IIT Hyderabad

- Under **Prof. Seshadri Shravan Kumar**: built CRNN OCR pipeline; compared with multimodal LLMs.

Teaching

IIT Hyderabad, Teaching Assistant, Convex Optimization — 2024, 2025

- Two semesters (Jan–May 2024, Jan–May 2025).

Technical Skills

Languages: Python, C++, C, SQL, JavaScript, Bash

Research tools: PyTorch, Hugging Face Transformers, LLM evaluation pipelines, linear probing, PCA, LaTeX

Engineering: Prompt engineering, retrieval and reranking, Ollama, Git, Docker